Session 3 - Linear models on feature vectors

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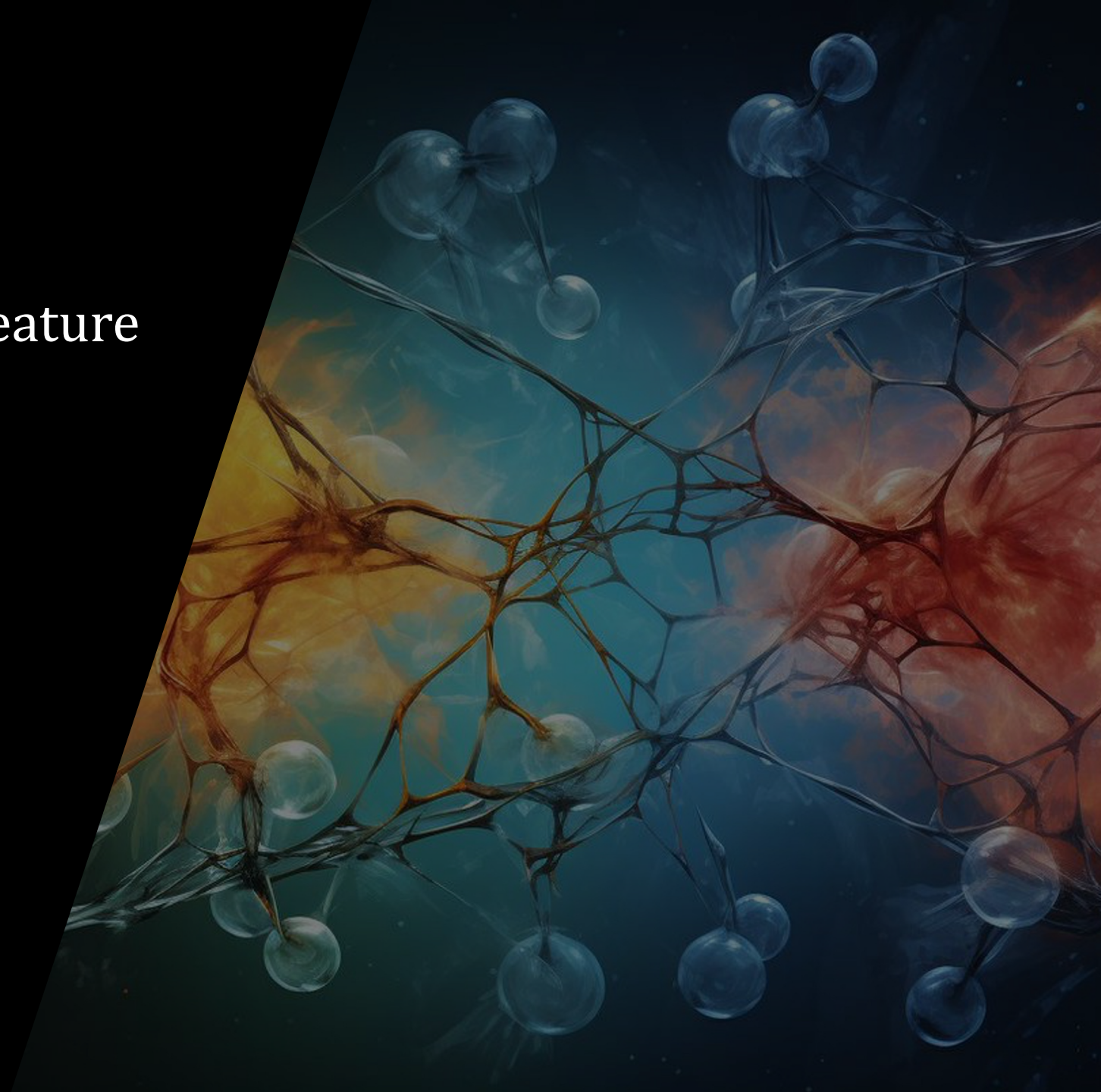


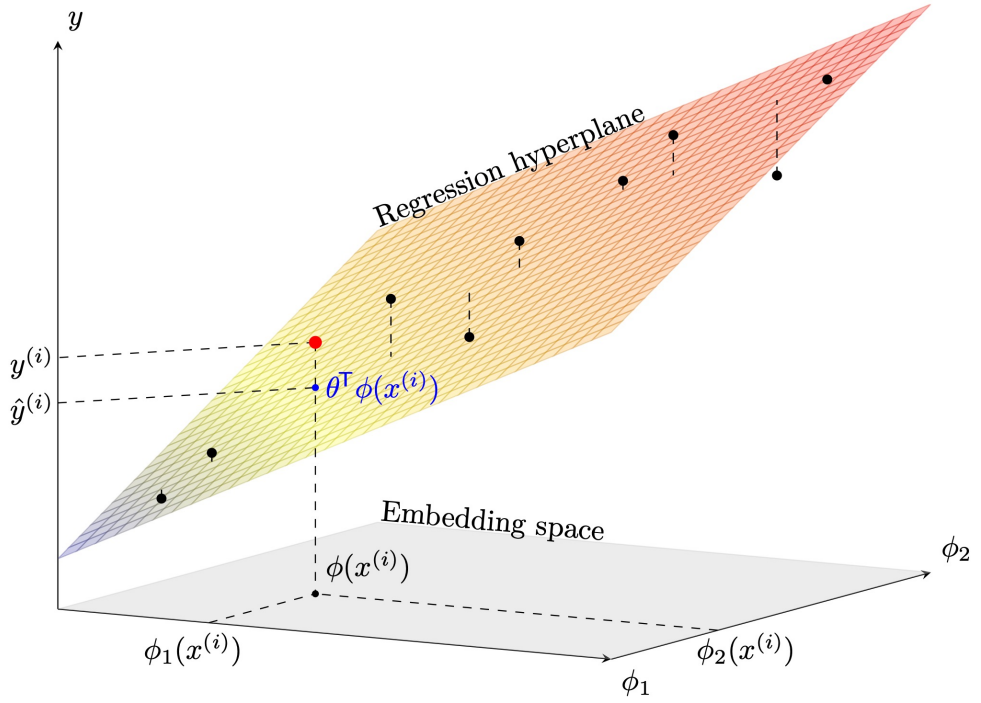
Linear regression: the model

Linear regression

- What kind of problems one can solve efficiently, even in large dimensions? → **Linear systems!**
- Let us talk first about regression: the answer is modelled as

$$f_{\boldsymbol{\theta}}\left(\phi_1^{(i)}, \phi_2^{(i)}, \cdots\right) = \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\phi}^{(i)} = \hat{y}^{(i)}$$

- It is sometimes convenient to add an affine term (also called **bias** in the neural network literature), which can be absorbed in the feature vector making it of dimension $d' + 1$ where $\boldsymbol{\theta} = [\theta_0, \theta_1, \theta_2, \cdots]^{\mathrm{T}}$, $\boldsymbol{\phi}^{(i)} = \left[1, \phi_1^{(i)}, \phi_2^{(i)}, \cdots\right]^{\mathrm{T}}$.
- **Geometric interpretation**: projection of an embedding vector onto a **hyperplane** parameterised by $\boldsymbol{\theta}$.



Linear regression: example and ERM

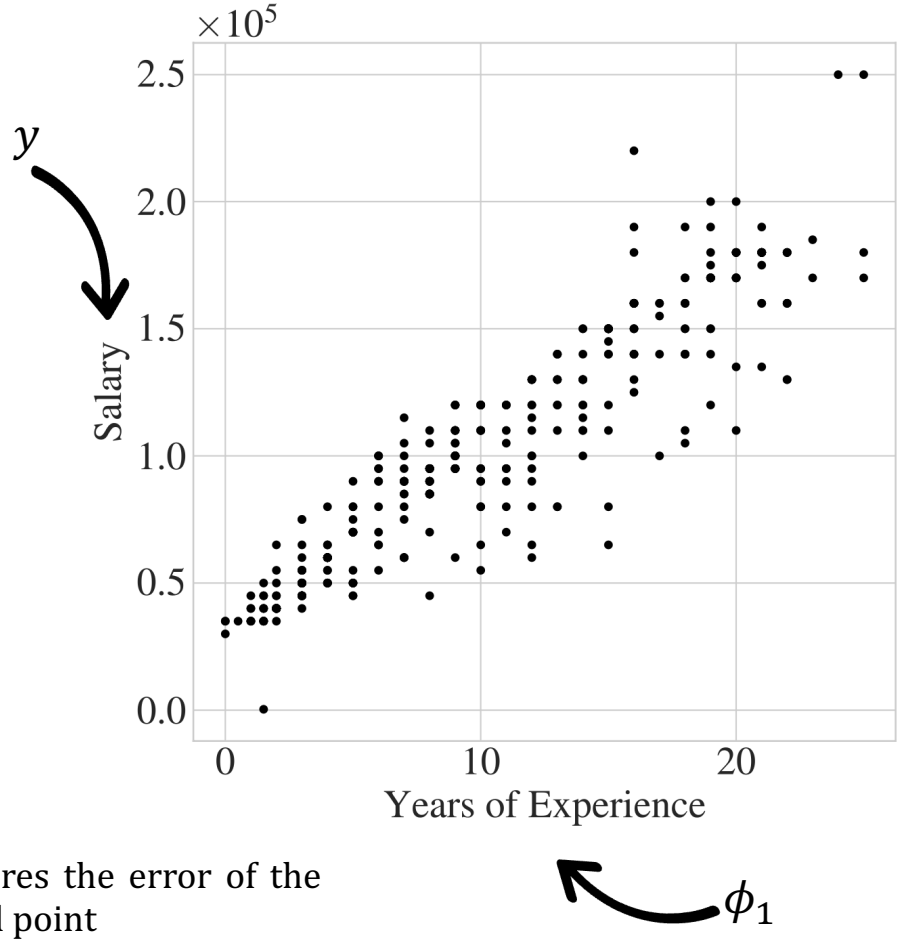
Linear regression

- Example: salary prediction based on the years of experience
- Data are 373 couples $\left(\phi^{(i)}, y^{(i)}\right) \Longrightarrow$ **Supervised learning**
- The target variable $y \in \mathbb{R}$ is continuous $\Longrightarrow$ **Regression**
- The linear model is

$$\hat{y}^{(i)} = \theta_0 + \theta_1 \phi_1^{(i)},$$

where $\phi_1^{(i)}$ is the nb. of years of experience of the i^{th} training example

- Now the model is fixed, how to find $\widehat{\boldsymbol{\theta}}$, the best possible parameters for our model and data?
- This is done using **empirical risk minimisation (ERM)**



$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, R(\boldsymbol{X}, \boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, \frac{1}{n} \sum_{i=1}^{n} \ell\left(\hat{y}^{(i)}, y^{(i)}\right)$$

Loss function: measures the error of the model on an individual point

Cost (or error) function: measures the error of the model on the training set

Linear regression: solution to ERM

Linear regression

- A common **choice** of loss for regression is a **squared loss function**

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

- ***Here***, the optimisation problem can be solved analytically in closed-form. Rewriting the risk matricially, we have

$$R(\boldsymbol{X}, \boldsymbol{\theta}) = \frac{1}{n}\|\boldsymbol{\Phi\theta} - \boldsymbol{y}\|_2^2$$

Feature matrix

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_1^{(1)} & \cdots & \phi_{d'}^{(1)} \\ \vdots & \ddots & \vdots \\ \phi_1^{(n)} & \cdots & \phi_{d'}^{(n)} \end{pmatrix} \in \mathbb{R}^{n \times d'}$$

Target vector

$$\boldsymbol{y} = [y_1, \ldots, y_n]^{\mathrm{T}} \in \mathbb{R}^n$$

The analytical minimisation of the squared loss in linear regression gives the unique solution (when $d' < n$) known as ***normal equations***

$$\widehat{\boldsymbol{\theta}} = \left(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi}\right)^{-1}\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{y}$$

Linear regression: illustration in 1D

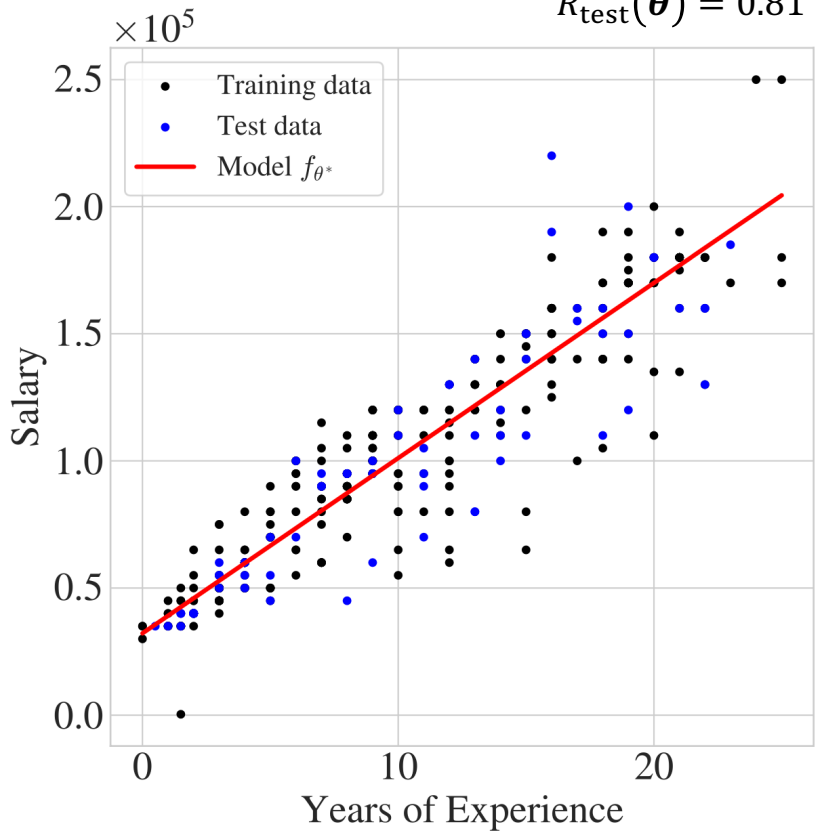
Linear regression

1. I **first** separated the dataset into **training and test sets**, $n = 298$ and $n_{\text{test}} = 75$ chosen randomly.
2. Then, I computed the optimal parameters minimising the empirical risk using the normal equations on the training features

$$\widehat{\boldsymbol{\theta}} = \left(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi}\right)^{-1}\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{y}.$$

3. I computed the risk on the train and test sets and found they are close.

$R_{\text{train}}(\widehat{\boldsymbol{\theta}}) = 0.87$
$R_{\text{test}}(\widehat{\boldsymbol{\theta}}) = 0.81$



\+ Exactly solvable model, low variance

\− Cannot represent local relationships, may be biased

Linear classification: the model

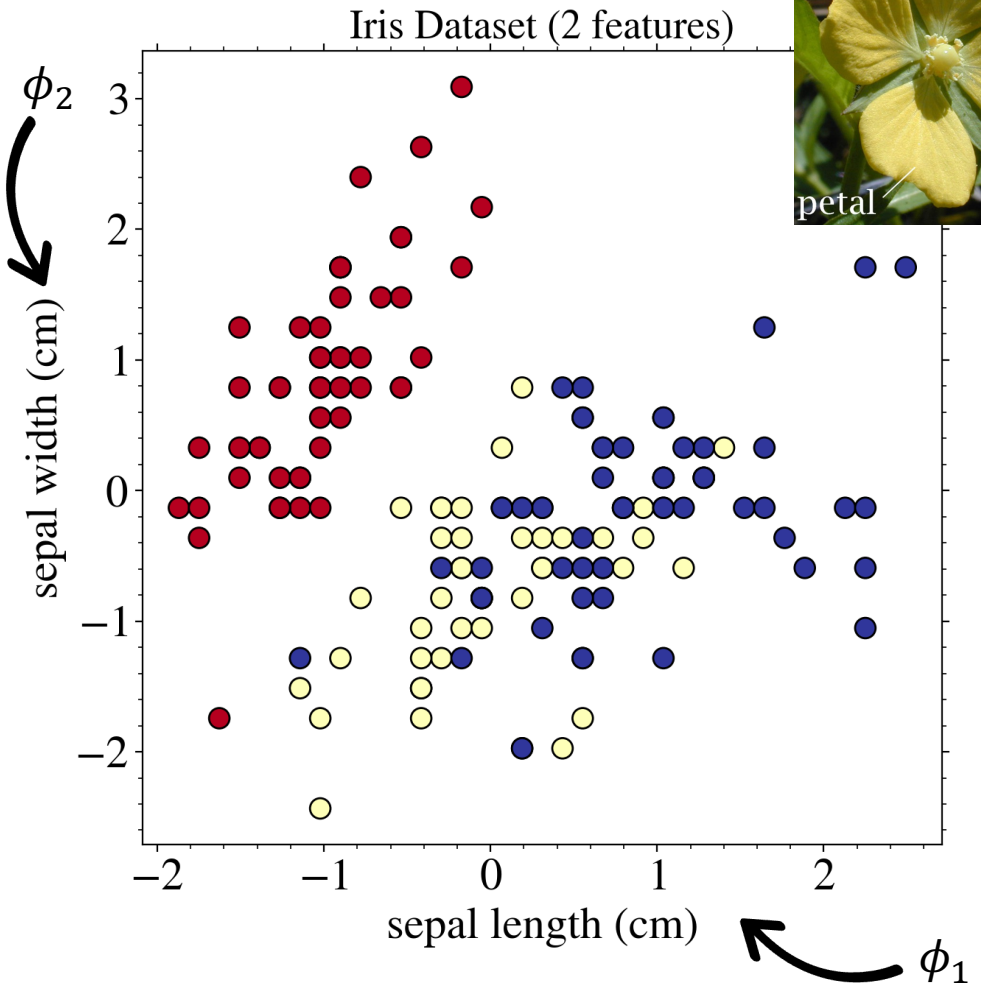
Linear classification

- Consider a K-class classification problem for which features $\boldsymbol{\phi}$ allow linear separability
- Example: Iris dataset with 150 couples $(\phi^{(i)}, y^{(i)}) \Longrightarrow$ **Supervised learning**
- The target variable $y \in \{0,1,2\} \Longrightarrow$ **Classification**
- A natural loss function for classification is the **0-1 loss**

$$\ell(y, \hat{y}) = \begin{cases} 1 & \text{if } \hat{y}^{(i)} \neq y^{(i)}, \\ 0 & \text{otherwise.} \end{cases}$$

- The optimal decision rule (in Bayes sense) is

$$\hat{y} = \text{argmax}_k \, p(y = k|\boldsymbol{\phi}).$$



We thus need a **model** $p_{\boldsymbol{\theta}}(y = k|\boldsymbol{\phi})$ of the conditional probability distribution to perform classification!

Linear classification

- The simplest models assume a linear log probability

$$\log p_\theta\left(y^{(i)} = k \middle| \boldsymbol{\phi}^{(i)}\right) = \boldsymbol{\theta}_k^{\mathrm{T}} \boldsymbol{\phi}^{(i)} - \log Z$$

where Z is a normalizing constant so that probabilities sum to one.

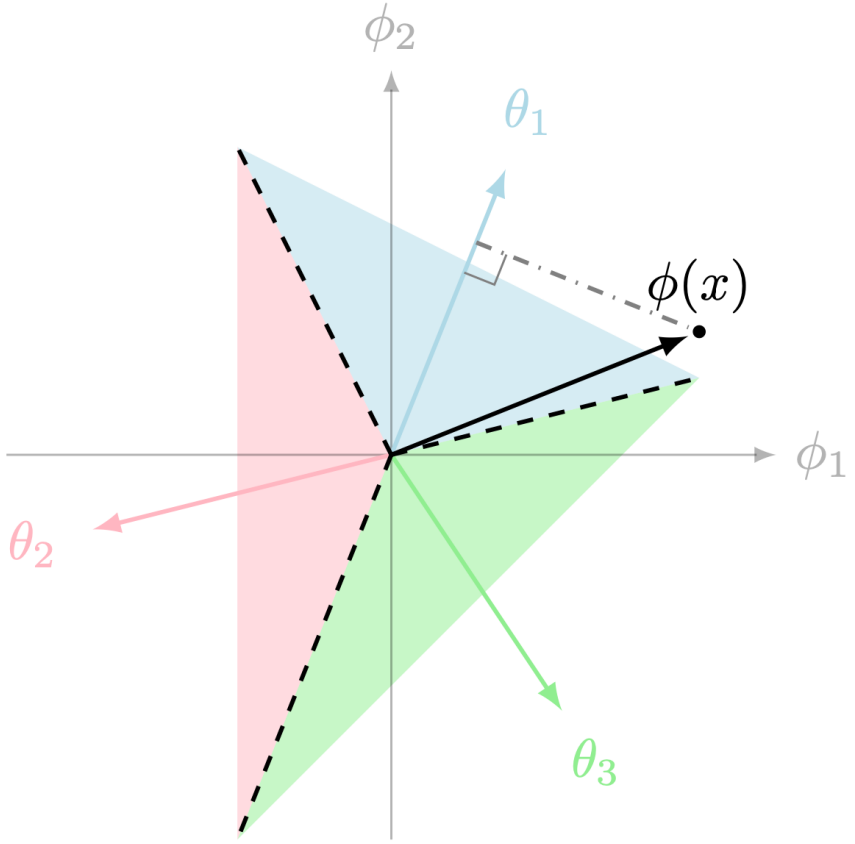
- It means that

$$p_\theta\left(y^{(i)} = k \middle| \boldsymbol{\phi}^{(i)}\right) = \frac{\exp\left(\boldsymbol{\theta}_k^{\mathrm{T}} \boldsymbol{\phi}^{(i)}\right)}{\sum_{j=1}^{K} \exp\left(\boldsymbol{\theta}_j^{\mathrm{T}} \boldsymbol{\phi}^{(i)}\right)}$$

which is called the **softmax function** allowing to turn the linear responses for each class into probabilities.

- And the classification rule is

$$\hat{y} = \mathrm{argmax}_k \, \boldsymbol{\theta}_k^{\mathrm{T}} \boldsymbol{\phi}^{(i)}$$



Geometrically, it corresponds to computing the overlap between the feature $\boldsymbol{\phi}^{(i)}$ and a vector representative for each class, $\boldsymbol{\theta}_k$, and associating **the class maximising the dot product**, leading to **linear decision** boundaries shown as hyperplanes.